

SESSION 17 — Modules and Packages

In Python, we often need functionality that someone else has already written.

Instead of writing everything from scratch, we can **reuse existing code** using **modules** and **packages**.

What is a Module

A **module** is simply a Python file that contains functions, variables, or classes that we can reuse in other programs.

In simple terms:

A module = a Python file with useful code.

Examples of Python modules

- `math.py`
- `statistics.py`
- `json.py`

These modules already exist in Python and provide many built-in functions.

What is a Package

A **package** is a collection of multiple modules grouped together.

Packages help organize large projects by grouping related modules.

Example:

The **pandas** package contains many modules used for data analysis.

Simple Summary

Concept	Meaning
Module	A single Python file containing reusable code
Package	A collection of multiple modules

Modules and packages help us **reuse code and avoid rewriting the same functionality**.

✓ Importing Modules

Python provides many built-in modules.

To use them we write

```
import module_name
```

Example

```
import math
```

```
# Using the math module

import math

print("Square root:", math.sqrt(16))
print("Value of pi:", math.pi)
print("Power:", math.pow(2, 3))
```

```
Square root: 4.0
Value of pi: 3.141592653589793
Power: 8.0
```

✓ Other Useful Built-in Modules

os

Used for interacting with the operating system.

datetime

Used for working with date and time.

json

Used for reading and writing JSON files.

statistics

Used for statistical calculations.

```
# datetime example

import datetime

now = datetime.datetime.now()

print("Current date and time:", now)
```

```
Current date and time: 2026-03-10 08:48:14.511133
```

```
# statistics example

import statistics

data = [10, 20, 30, 40]

print("Mean:", statistics.mean(data))
print("Median:", statistics.median(data))
```

```
Mean: 25
Median: 25.0
```

✓ Creating a Custom Module

We can create our own module.

Step 1

Create a Python file called utils.py

Step 2

Write functions inside it.

Example

```
# Example content of utils.py

def greet(name):
    print("Hello", name)

def add(a, b):
    return a + b
```

Double-click (or enter) to edit

Installing Packages using pip

pip is the Python package manager.

Example

- pip install pandas
- pip install numpy
- pip install matplotlib

SESSION 18 — REGULAR EXPRESSIONS

What is Regular Expression (Regex)

Regex is used for pattern matching in text.

Example tasks

Find email addresses in text

Find phone numbers

Extract dates

Clean messy text data

Python uses the module

re

✓ Common Regex Patterns

Email

- \S+@\S+

Phone number

- \d{10}

Date

- `\d{2}/\d{2}/\d{4}`

Number

- `\d+`

```
import re

text = "My email is student@gmail.com sahil@gm ail.com"

emails = re.findall(r"\S+@\S+", text)

print(emails)

['student@gmail.com', 'sahil@gm']
```

```
# re.search example

import re

text = "Order number is 123"

match = re.search(r"\d+", text)

print(match.group())

123
```

```
# re.sub example

import re

text = "Price is 100 dollars"

clean = re.sub(r"\d+", "XXX", text)

print(clean)

Price is XXX dollars
```

```
import re

text = """
Contact us at support@gmail.com
or admin@yahoo.com
"""

emails = re.findall(r"\S+@\S+", text)

print("Emails found:", emails)

Emails found: ['support@gmail.com', 'admin@yahoo.com']
```

✓ Practice Questions

1. Import the math module and print the square root of 49.

2. Use the statistics module to calculate the mean of the list [5, 10, 15].
3. Use regex to extract all numbers from the text
"The price is 50 and discount is 10".
4. Write a regex pattern to extract all email addresses from a given text.

Start coding or [generate](#) with AI.